CS5242 Project

Image Classification on Mini-ImageNet

Frozen Features, Fine-Tuning, and From Scratch — a Three-Lens Comparison

Agenda

1. **Problem, Data & Pre-processing** — Mini-ImageNet, EDA, normalisation choices
2. **Three Approaches** — baseline + proposed improvement
3. **Approach 1** — Classical ML on frozen pretrained features
4. **Approach 2** — Fine-tuning with selective unfreezing
5. **Approach 3** — Training from scratch
6. **Conclusions** — cross-approach synthesis

Part 1

Problem, Data & Pre-processing

Problem Statement & Motivation

Task. 100-class image classification on **Mini-ImageNet** — small enough to iterate on a single GPU yet rich enough to distinguish modern architectures.

Why this problem? Canonical CV task where trade-offs between **architecture**, **data regime**, and **pretraining** can be measured cleanly. Mini-ImageNet stresses **small data** (500/class), **low resolution** (32×32), and the **pretrained-vs-scratch** divide.

Why deep learning? 100-way image classification needs hierarchical visual features — CNN/ConvNeXt inductive biases match the input structure; no competitive classical alternative at this scale.

Why pretrained models — justified. We use them in **A1 & A2** to quantify how much Mini-ImageNet performance is carried by ImageNet pretraining, and include **A3 (from scratch)** to isolate that contribution. The scientific question is *the value of pretraining*.

Dataset

- **Source:** Mini-ImageNet [4] — 100 classes drawn from ImageNet-1K
- **Splits:** 50,000 train / 10,000 val / 5,000 test (fixed, reproducible)
- **Balance:** 500 train images per class (perfectly balanced — no re-weighting needed)
- **Resolutions tested:** native (≈500×N) resized to **224×224** and **32×32**

Why 32×32 *and* 224×224?

- **224×224** matches pretraining → upper-bound on pretrained feature quality
- **32×32** stresses architectures *off* design regime — robustness test + practical memory-bandwidth setting

Known Caveat

Mini-ImageNet's 100 classes are a *subset* of ImageNet-1K. Pretrained backbones have seen these classes, so 224×224 pretrained numbers are an **upper bound**, not transfer to an unseen distribution. Approach 3 (from scratch) addresses this directly.

Exploratory Data Analysis

Class distribution

- Exactly balanced across train/val/test — no class imbalance to correct
- Eliminates class-weighting as a confound in later results

Dataset statistics

- **Train mean (RGB):** `(0.479, 0.452, 0.408)`
- **Train std (RGB):** `(0.288, 0.280, 0.294)`
- Native resolutions cluster near 500×375

Visual sample (random grid)

> EDA outputs in experiments/eda/ — full figures from `data_analysis.ipynb`.

Pre-processing Pipeline

Shared across all approaches

- **Resize** to target (224 or 32) with aspect-preserving centre crop
- **Normalise** with this dataset's own mean/std (from the train split), not ImageNet defaults
- **Channel order** RGB, tensor in `[0,1]` before normalisation
- Split integrity: transforms fit on **train only**; val/test use those stats

Approach-specific

- **A2 & A3 (training):** random crop + horizontal flip augmentation
- **A1 (frozen features):** no train-time augmentation — features extracted once and cached
- **A3 (from scratch):** heavier augmentation may be used — *[team member to specify]*

> Computing mean/std *on this dataset* is deliberate: the 32×32 resized distribution's statistics drift from full-resolution ImageNet statistics.

Part 2

Three Approaches — Baseline & Improvement

The Three-Lens Design

Approach 1 — Classical ML

Role: strong baseline using **pretrained** features

- Frozen backbone + linear SVM / LogReg
- Isolates feature quality from classifier choice
- Very cheap, interpretable

Approach 2 — Fine-Tuning

Role: proposed improvement

- Pretrained init + selective unfreezing
- Combines transfer with task-specific adaptation
- Expected Pareto-best at moderate cost

Approach 3 — Scratch

Role: baseline from **first principles**

- Random init, full training
- Same architecture family, no ImageNet prior
- Measures *pure* in-domain signal

> **Scientific question:** how much of Mini-ImageNet performance is *feature quality from pretraining* (A1) vs *task-specific adaptation* (A2) vs *learned end-to-end without prior* (A3)?
> Reporting all three lets us attribute performance to each source instead of conflating them.

Part 3

Approach 1 — Classical ML

Pretrained Backbones as Frozen Feature Extractors

(Strong baseline using pretrained features)

Approach 1: Motivation

Idea: Use pretrained ImageNet backbones as **frozen feature extractors**, then train lightweight classical classifiers on the extracted features.

Advantages

- **Data efficient** — only classifier head is learnable
- **Fast** — single forward pass + seconds-to-minutes training
- **Interpretable** — isolates backbone effect

Limitations

- No task-specific feature adaptation
- Resolution sensitivity (pretrained for 224×224)
- Linear classifier ceiling

Approach 1: Pipeline

```
Image → [Pretrained Backbone (frozen)] → Global Avg Pool → Feature Vector → [SVM / LogReg] → Class
```

Pipeline:

1. Load pretrained ImageNet backbone — freeze all weights
2. Extract feature vectors via forward pass (one-time cost)
3. Train classical classifier on extracted features

Classifiers (sklearn, default hyperparameters):

- **Linear SVM** — `LinearSVC`, **squared-hinge** loss + L2, OvR, liblinear coordinate-descent solver
- **Logistic Regression** — multinomial cross-entropy + L2, LBFGS, `max_iter=2000`

> Results are single-seed with default hyperparameters.

Approach 1: Experimental Design

Backbones (3 architecture families)

Backbone	Family	Params (backbone, M)	Feature Dim	torchvision weights (IN-1K top-1)
ConvNeXt-Tiny	ConvNeXt [2]	27.9	768	`IMAGENET1K_V1` (82.52%)
ResNet-18	ResNet [1]	11.2	512	`IMAGENET1K_V1` (69.76%)
ResNet-34	ResNet [1]	21.3	512	`IMAGENET1K_V1` (73.31%)
ResNet-50	ResNet [1]	23.7	2048	`IMAGENET1K_V2` (80.86%)
EfficientNet-b0	EfficientNet [3]	4.1	1280	`IMAGENET1K_V1` (77.69%)

Experiment Grid

- **32×32**: All backbones × {SVM, LogReg}
- **224×224**: ConvNeXt-Tiny & ResNet-18 × SVM (downselected)

[1] He et al., CVPR 2016 [2] Liu et al., CVPR 2022 [3] Tan & Le, ICML 2019 [4] Vinyals et al., NeurIPS 2016

Approach 1 — Results: NetScore, Accuracy, Inference & Parameters

> **Key observation:** ConvNeXt-Tiny leads on **accuracy** (44.16%) despite the most parameters, while ResNet-18 leads on **NetScore** (45.4, per Wong [6]) due to its small size and fast inference. EfficientNet-b0 is slowest and least accurate.

Approach 1 — Efficiency Comparison (32×32)

$$\text{NetScore} = 20\,\log_{10}\left(\frac{A^2}{\sqrt{T}\,\sqrt{P}}\right)$$

Backbone	Clf	Acc	Infer	NS
ResNet-18	SVM	32.52	**2.89**	**45.4**
ResNet-18	LR	32.46	3.09	45.1
ConvNeXt	LR	**44.16**	5.92	43.6
ConvNeXt	SVM	43.28	5.96	43.2
ResNet-34	LR	33.16	5.17	40.4
ResNet-34	SVM	33.08	5.49	40.1
EffNet-b0	LR	24.72	9.66	39.7
EffNet-b0	SVM	23.88	9.51	39.2
ResNet-50	LR	33.22	6.47	39.0
ResNet-50	SVM	27.34	6.51	35.6

Key Observations

- **ResNet-18 SVM tops NetScore** (45.4) — smallest model (11.2M) + fastest inference (2.89 ms) offsets lower accuracy
- **ConvNeXt-Tiny LogReg** leads on raw accuracy (44.16%) but 27.9M params penalise its NetScore
- **Classifier choice barely affects NetScore** — SVM vs LogReg differ by <0.5 for same backbone, since inference time is backbone-dominated
- **EfficientNet-b0** has poor NetScore despite fewest params (4.1M) — slowest inference (9.5 ms) + lowest accuracy (24%)

Why ConvNeXt-Tiny Dominates at 32×32

The key lies in how each architecture's **stem** processes low-resolution input:

ConvNeXt-Tiny Stem

- **4×4 stride-4 patchify** convolution
- 32×32 input → **8×8 feature map** after stem
- Subsequent **7×7 depthwise kernels** still have meaningful spatial extent to work with
- LayerNorm stabilises activations; GELU preserves gradient flow

ResNet Stem

- **7×7 stride-2** conv + **3×3 stride-2** max-pool
- 32×32 input → **8×8** after stem, then quickly **1×1** through residual stages
- Deeper layers receive **spatially degenerate** feature maps — no local structure to exploit
- BatchNorm statistics are calibrated for 224×224 distributions

> ConvNeXt's patchify stem is resolution-adaptive: it reduces spatial dims in one step without the cascading downsampling that collapses small inputs in ResNets.

Why Architecture Matters More Than Depth

- **ResNet-18/34/50 cluster within ~1% on LogReg** (32.46 / 33.16 / 33.22) → extra depth gives no headroom at 32×32
- **ConvNeXt-Tiny** leads by 10+% — patchify stem + 7×7 DW-Conv preserve spatial structure
- **EfficientNet-b0 underperforms** — designed for 224×224; at 32×32 its feature maps are too small for Squeeze-and-Excitation (SE) attention and depthwise convolutions to be effective
- *Caveat:* RN-50 uses torchvision `V2` weights; the "depth doesn't help" claim holds for LogReg but SVM+RN-50 is an outlier

Generalisation Gap

> Val–Test gap is uniformly small (0.6–3.0%) — linear models on frozen features generalise well.

Resolution Impact & t-SNE Feature Visualisation

- Both backbones lose ≈**50%** (224→32) — severe resolution floor
- ConvNeXt-Tiny's ~10% advantage preserved at both resolutions

- **224×224:** Tight, well-separated clusters — high linear-probe accuracy
- **32×32:** Clusters collapse — consistent with ~50% drop
- ConvNeXt-Tiny retains more inter-class separation at both resolutions

Backbone Downselection for Later Approaches

Based on these results, we select **two backbones** for Approaches 2 and 3:

1. ConvNeXt-Tiny (SOTA)

- **Highest accuracy** at all resolutions
- 93.88% (224×224), 44.16% (32×32)
- Modern architecture innovations
- Best candidate for further improvement

2. ResNet-18 (Classical Baseline)

- **Fastest inference** (2.9 ms/image)
- Well-established architecture
- Measures how much fine-tuning closes the gap
- Simple & interpretable

> This pairing lets us compare fine-tuning and from-scratch strategies on a **modern vs classical** architecture.

Approach 1 — Key Takeaways

1. **ConvNeXt-Tiny is the strongest backbone** — leads by 10+% at both resolutions on Mini-ImageNet
2. **Architecture matters more than depth** — RN-18/34/50 cluster within ~1%; ConvNeXt's stem design is the larger effect
3. **Resolution is the dominant factor** — 224→32 costs ≈50% for both probed backbones; larger than any architecture gap
4. **EfficientNet-b0 underperforms at 32×32** — designed for 224×224; feature maps too small for SE attention and depthwise convolutions
5. **Small val–test gap (0.6–3.0%)** — linear probes on frozen features generalise well
6. **Limitations.** Single seed, default hyperparameters, Mini-ImageNet ⊂ ImageNet-1K, mixed `V1` / `V2` torchvision weights

Part 4

Approach 2 — Fine-Tuning

Approach 2: Motivation & Setup

Idea: Start from ImageNet-pretrained ConvNeXt-Tiny and ResNet-18, then adapt the classifier, the deepest stage, the full model, or low-rank adapter paths to Mini-ImageNet.

Why this improves Approach 1

- Approach 1 freezes the representation and only learns a linear boundary
- Fine-tuning lets high-level features reorganise around Mini-ImageNet's 100 classes
- Tests whether ConvNeXt's frozen-feature advantage survives task adaptation

Shared protocol

- **Input:** 32×32 Mini-ImageNet
- **Backbones:** ConvNeXt-Tiny, ResNet-18
- **Runs:** `USE_AUG=False`, `mix_mode=none`
- **Optimiser:** AdamW [8], LR = `1e−4`
- **Schedule:** cosine annealing [9] + early stopping

> Fine-tuning asks: how much can we recover from the 32×32 resolution floor once the backbone is allowed to adapt?

Approach 2: Four Adaptation Policies

Policy	What is trainable?	Patience	ConvNeXt trainable	ResNet trainable
Classifier only (`backbone`)	final classifier head only	3	0.078M	0.051M
Last stage + classifier	deepest feature stage + classifier	7	15.549M	8.445M
Full fine-tuning (`none`)	all pretrained weights + classifier	7	27.897M	11.228M
LoRA [10]	Linear-layer adapters + classifier; base weights frozen	5	0.608M	0.056M

Why these policies?

- **Classifier only** is the transfer-learning analogue of the Approach 1 frozen-feature baseline, but with a neural head
- **Last stage** targets class-specific semantic features while preserving generic early filters
- **Full fine-tuning** gives maximum adaptation capacity
- **LoRA** tests parameter-efficient adaptation on eligible `Linear` layers only, with rank = 8 and alpha = 16

> These policies move from cheapest and most constrained to most flexible, letting us separate classifier learning from representation adaptation.

LoRA: Low-Rank Adaptation

LoRA is applied only to eligible backbone `Linear` layers in our implementation; convolutional layers are not LoRA-adapted.

$$W' = W_{frozen} + BA\frac{\alpha}{r}$$

What is trained?

- LoRA matrices `A` and `B` in targeted `Linear` layers
- Final classifier head
- Original backbone weights remain frozen
- Rank `r = 8`, scaling `alpha = 16`

Why it matters here

- ConvNeXt has many internal `Linear` layers, so LoRA can adapt useful representation paths
- ResNet-18 is mostly convolutional, so Linear-layer LoRA has less room to help
- Explains the architecture-specific LoRA result: **61.74%** for ConvNeXt vs **19.64%** for ResNet-18

> LoRA is parameter-efficient, but its benefit depends on whether the backbone exposes useful `Linear` layers for low-rank updates.

AdamW Optimizer

All transfer-learning runs use AdamW [8], an Adam-style optimizer with decoupled weight decay:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2$$

$$\theta_t = \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} - \eta \lambda \theta_{t-1}$$

Why AdamW?

- Adaptive step size for each parameter
- Weight decay is applied separately from gradient moments
- Stable for fine-tuning pretrained ConvNeXt and ResNet at LR = `1e-4`

Parameters

- `g_t`: gradient at step `t`
- `m_t`: moving average of gradients; `beta_1` controls momentum
- `v_t`: moving average of squared gradients; `beta_2` controls gradient scale memory
- `eta`: learning rate, `lambda`: weight decay, `epsilon`: stability constant

> Compared with SGD, AdamW is less sensitive to one global learning rate and handles layers with different gradient scales better.

Cosine Annealing Scheduler

All fine-tuning runs use AdamW with a cosine learning-rate schedule:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min})\left(1 + \cos\left(\frac{t}{T_{max}}\pi\right)\right)$$

Why use it?

- Larger updates early help the model move toward Mini-ImageNet-specific features
- Smaller updates later avoid disrupting useful pretrained representations
- Smooth decay is better suited to fine-tuning than abrupt LR drops

In our setup

- Initial LR: `1e−4`
- Schedule length: full epoch budget
- Checkpoint selection: best validation accuracy
- Early stopping prevents wasting epochs after validation saturation

> Cosine annealing gives adaptation room early, then stabilises the pretrained backbone as training converges.

Approach 2 — Accuracy Results

Backbone	Method	Best val	Test acc	Test loss	Epochs
ConvNeXt-Tiny	Classifier only	52.40	50.92	1.9369	54
ConvNeXt-Tiny	Last stage + classifier	57.26	55.44	1.7156	16
ConvNeXt-Tiny	**Full fine-tuning**	**65.11**	**62.70**	1.6357	16
ConvNeXt-Tiny	LoRA	64.26	61.74	**1.4397**	34
ResNet-18	Classifier only	21.49	19.98	3.5373	36
ResNet-18	Last stage + classifier	33.18	32.64	2.8840	18
ResNet-18	**Full fine-tuning**	**40.41**	**39.34**	**2.5783**	15
ResNet-18	LoRA	21.48	19.64	3.5176	40

> Full fine-tuning is best for both backbones; ConvNeXt-Tiny LoRA is almost as accurate and has the lowest ConvNeXt test loss.

Approach 2 — What the Results Say

ConvNeXt-Tiny

- Classifier only already improves over the Approach 1 frozen LogReg baseline: **50.92% vs 44.16%**
- Last-stage fine-tuning adds another **+4.52%**
- Full fine-tuning reaches **62.70%**, the strongest result
- LoRA reaches **61.74%**, only **0.96%** behind full tuning

ResNet-18

- Classifier-only transfer is weak: **19.98%**, below the Approach 1 frozen-feature probes
- Last-stage tuning recovers to **32.64%**
- Full fine-tuning reaches **39.34%**, best ResNet result
- LoRA stays near classifier-only because adapters only target `Linear` layers

> ConvNeXt-Tiny benefits from both full fine-tuning and LoRA because its ConvNeXt blocks contain many internal Linear layers; ResNet-18 needs broader convolutional adaptation.

Approach 2 — Efficiency & NetScore

Backbone	Method	Test	Infer	Params	Train time	Peak mem	NetScore
ConvNeXt	**Full fine-tuning**	**62.70**	6.01	27.90M	1950s	1458MB	**49.65**
ConvNeXt	Last stage	55.44	5.90	27.90M	**1872s**	437MB	47.59
ConvNeXt	LoRA	61.74	10.58	28.43M	3986s	1126MB	46.84
ConvNeXt	Classifier only	50.92	6.04	27.90M	6237s	**260MB**	46.01
ResNet-18	**Full fine-tuning**	**39.34**	2.99	11.23M	**1773s**	513MB	**48.53**
ResNet-18	Last stage	32.64	2.89	11.23M	2116s	284MB	45.44
ResNet-18	Classifier only	19.98	**2.80**	11.23M	4154s	187MB	37.05
ResNet-18	LoRA	19.64	2.94	11.23M	4569s	**187MB**	36.54

> Full fine-tuning wins NetScore for both backbones because the accuracy gain outweighs the inference-time and parameter terms in the metric.

Approach 2 — Transfer Learning Takeaways

Main ranking

1. **ConvNeXt full fine-tuning:** 62.70%
2. **ConvNeXt LoRA:** 61.74%
3. **ConvNeXt last-stage:** 55.44%
4. **ConvNeXt classifier-only:** 50.92%

Cross-backbone lesson

- ConvNeXt-Tiny beats ResNet-18 under every transfer policy
- Full tuning improves both architectures most reliably
- LoRA is architecture-dependent: strong on ConvNeXt, weak on ResNet-18
- More trainable capacity improves accuracy, but raises overfitting and memory risk

> The proposed improvement is supported: task-specific adaptation raises the best 32×32 result from **44.16%** frozen features to **62.70%** full fine-tuning.

Part 5

Approach 3 — Training from Scratch

Part 6

Conclusions

Conclusions

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Thank You